

EV Charging Demand Forecasting

Insight Summary

- **Charging demand follows a clear daily rhythm.**

Sessions are near zero overnight (midnight- 4 a.m.), rise sharply from 5 a.m., and peak between 10 a.m. and 2 p.m., consistent with daytime, workplace-style charging rather than overnight home charging.

- **Weekdays see more demand than weekends**

Average sessions per hour run about 12 on weekdays versus 8.8 on weekends, reinforcing that usage is driven by commuter/workday patterns.

- **No strong seasonal (month-to-month) patterns were found**

Unlike real-world EV charging data, monthly averages stayed within a narrow band year-round.

- **Most of the historical demand growth reflects network expansion, not rising demand per station**

Total sessions jumped sharply at the start of 2023 and 2024, matching almost exactly when new charging stations came online (8 to 17 to 33 stations), not a surge in per-station usage.

- **XGBoost (using calendar information) was the most accurate forecasting model**

It outperformed Prophet and SARIMA on nearly every metric, when all three were tested under identical, fair conditions.

- **The two forecasting models disagree on future growth, and this is a genuine unresolved certainly**

XGBoost assumes demand will hold flat at 2024 levels through 2025, while Prophet assumes the historical growth data continues, producing a forecast 6% higher.

- **For staffing and capacity planning**

Resourcing should be weighted toward late-morning through mid-afternoon on weekdays, with minimal overnight coverage needed. Given the disagreement between models on future growth, it may be prudent to plan capacity toward the higher (Prophet) estimate as a buffer, while treating the flatter (XGBoost) estimate as the more conservative baseline.

- **A significant data quality issue was identified and documented**

30% of all charging sessions occur before their charger's recorded installation date, a pattern affecting every charger in the dataset and correlating strongly with how late each charger came online. However, it does not affect this forecasting task, since installation data wasn't used as a model input.